Chapter 3

All excercises without a ★ from 3.1–3.13

Homework 3 due: Tuesday 18$^{\text{th}}$ March, 2014

Exercise 3.1. Prove that 0_R and R are ideals of R for any ring R. The ideal 0_R is called the *zero ideal*.

Proof. We already know that 0_R and R are non-empty subrings of R. The only element in 0_R is 0_R, and $a0_R = 0_Ra = 0_R$ for all $a \in R$. So 0_R is an ideal. Also, ar and ra are both in R for any $r, a \in R$. So R is an ideal as well. □

Exercise 3.2. Let $\mathbb{R}[x]$, the set of all polynomials with real coefficients. Let C be the subset of $\mathbb{R}[x]$ with constant terms that are 0. For example, $x^3 + x \in C$ but $x^3 + x - 4 \notin C$. Prove that C is or is not an ideal.

Proof. C is an ideal. $x \in C$ so then C is non-empty.

Let $i, j \in C$. $i - j \in C$ since both i and j have a constant term of 0, and so the constant term of $i - j$ is 0.

Any polynomial in $\mathbb{R}[x]$ can be written as $f(x)+c$ where c is the constant term and $f(x) \in C$. Let $f(x)+0 \in C$ and $g(x)+c \in \mathbb{R}[x]$. $f(x)(g(x)+c) = f(x)g(x)+f(x)c$ which has a constant term of 0. So then C is an ideal by the ideal test. □

Exercise 3.3. Prove that the set $I = \{(a, 0) \colon a \in \mathbb{Z}\}$ is an ideal of the ring $\mathbb{Z} \times \mathbb{Z}$.

Proof. $\mathbb{Z} \times \mathbb{Z} \neq \emptyset$ since $(1, 0) \in \mathbb{Z} \times \mathbb{Z}$. Let $(a, 0), (b, 0) \in \mathbb{Z} \times \mathbb{Z}$. $(a, 0) - (b, 0) = (a - b, 0) \in \mathbb{Z} \times \mathbb{Z}$. Now let $(x, y) \in \mathbb{Z} \times \mathbb{Z}$. $(x, y)(a, 0) = (x \cdot a, y \cdot 0) = (x \cdot a, 0) \in \mathbb{Z} \times \mathbb{Z}$. So $\mathbb{Z} \times \mathbb{Z}$ is an ideal by the ideal test. □

Exercise 3.4. Let A and B be ideals of R. Prove that $A \cap B$ is also an ideal of R.

Proof. A and B are both ideals. Thus $0_R \in A$ and $0_R \in B$ and $0_R \in A \cap B$ ($0_R \in I$ for any ideal in R. If not then $a \cdot 0_R = 0_R \notin I$ for any $a \in I$.)

Let $x, y \in A \cap B$. $x - y \in A$ and $x - y \in B$ since A and B are both subrings, and subrings are closed under subtraction.

Let $r \in R$. $rx, xr \in A$ and $ry, yr \in B$ since A and B are ideals. So each of these products are also in $A \cap B$, and $A \cap B$ is an ideal by the ideal test. □

Exercise 3.5. Let A and B be ideals of R. Prove that $A \cup B$ is not necessarily an ideal of R.

Proof. Consider the ring $\mathbb{Z}$ with ideals $2\mathbb{Z}$ and $3\mathbb{Z}$. $2 \in 2\mathbb{Z}$ and $3 \in 3\mathbb{Z}$, but $3-2 = 1 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$.

Remark. The union of plenty ideals *is* an ideal, e.g., $I \cup 0_R$ and $I \cup I$ for any ideal I in R. Recall that the union of subrings is not necessarily a subring.

□

Exercise 3.6. If I is an ideal in the ring R and J is an ideal of the ring S, prove that $I \times J$ is an ideal of the ring $R \times S$.

Proof. $I, J \neq \emptyset$ as they are both ideals. So there exists and $i \in I$ and $j \in J$ such that $(i, j) \in I \times J$, and $I \times J \neq \emptyset$.
Let $i, i' \in I$ and $j, j' \in J$. $(i, j) - (i', j') = (i - i', j - j') \in I \times J$ as $i - i' \in I$ and $j - j' \in J$ since I and J are both ideals.
Let $(r, s) \in R \times S$. $(i, j)(r, s) = (ir, js) \in R \times S$. Similarly $(r, s)(i, j) \in R \times S$. So $I \times J$ is an ideal by the ideal test. $\square$

Exercise 3.7. Let R be a ring with unity 1_R and let I be an ideal of R. If $1_R \in I$ prove $I = R$.

Proof. $I \subset R$ by definition of an ideal. If $1_R \in I$ then $1_R r$, $r 1_R \in I$ since I is an ideal. So then $1_R r = r 1_R = r \in I$ for all $r \in R$ which implies that $R \subset I$ and $R = I$. $\square$

Exercise 3.8. Let R be a ring with unity 1_R and let I be an ideal of R. If I has an element x and x^{-1} such that $x \cdot x^{-1} = x^{-1} \cdot x = 1_R$, prove that $I = R$.

Proof. So if I has two elements such that $x \cdot x^{-1} = x^{-1} \cdot x = 1_R$ then $1_R \in I$ since I is a subring and closed under multiplication. It follows that $I = R$ by the exercise above. $\square$

Exercise 3.9. ★

Exercise 3.10. ★

Exercise 3.11. ★

Exercise 3.12. Show that $5\mathbb{Z}$ is a prime ideal.

Proof. We know from example 3.1 that $5\mathbb{Z}$ is an ideal in $\mathbb{Z}$. $6 \notin 5\mathbb{Z}$, so it is a proper ideal. Let $a, b \in \mathbb{Z}$ such that $ab \in 5\mathbb{Z}$. So then $ab = 5n$ for some $n \in \mathbb{Z}$ which implies that $5|ab$. Suppose that $b \notin 5\mathbb{Z}$ so that $5 \nmid b$. Since 5 is prime, $5|a$ (by Euclid's Lemma). Thus $5\mathbb{Z}$ is a prime ideal. $\square$